Práctica 2: Sistema Cardiovascular

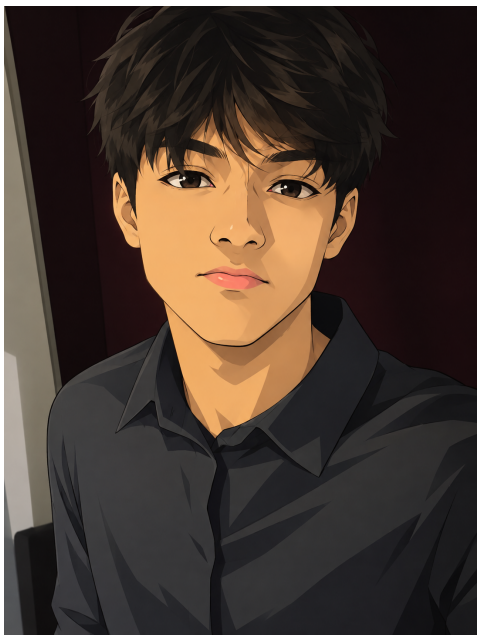
Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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Datos de la simulación

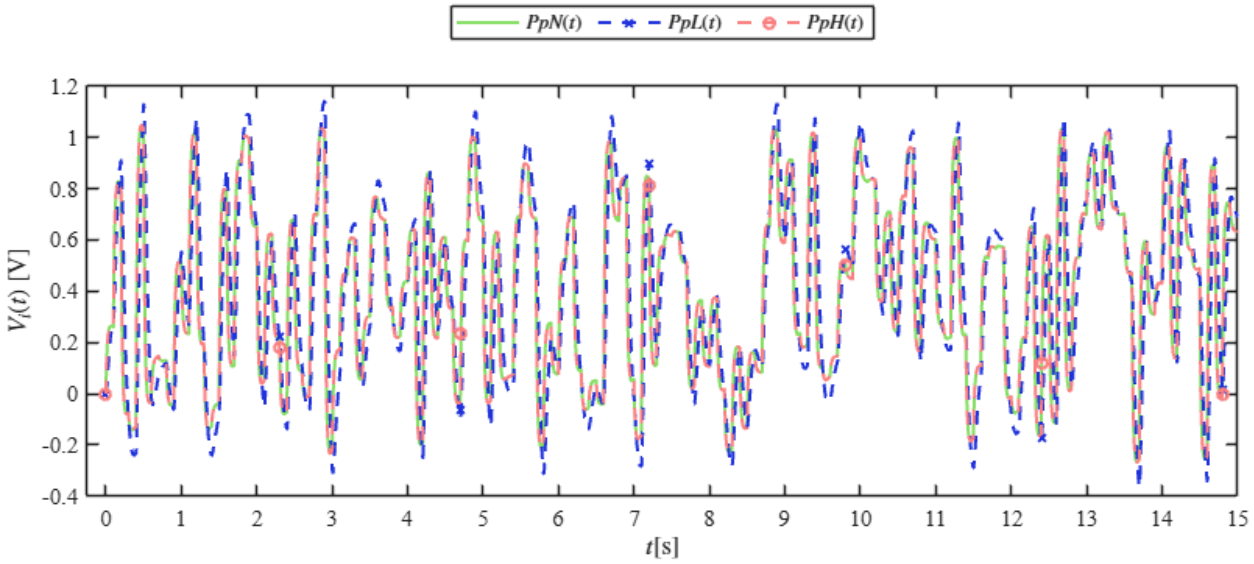
```
clc; clear; close all; warning('off','all')
open_system('Sistema2')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';

x = sim('Sistema2.slx',parameters);
t_data = (0:0.1:15)';
t = (0:0.001:15)';

Pa = interp1(t_data,x.Pa, t, 'previous');
writematrix(Pa,'signal.xlsx');
```

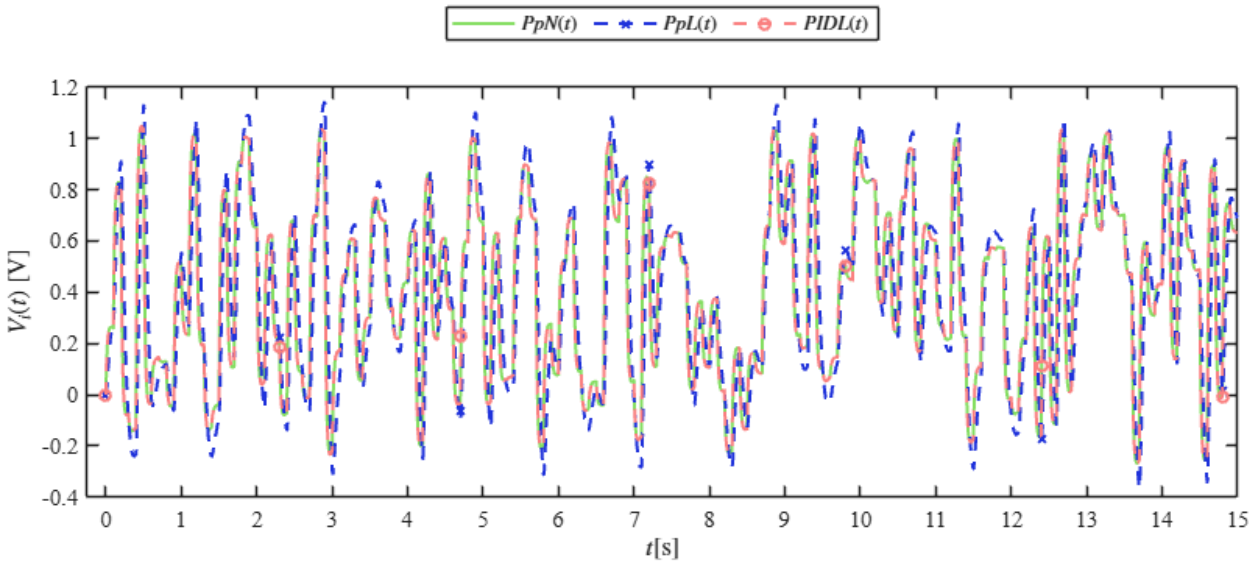
Respuesta a los lazos abiertos

```
plotsignals(x.t,x.PpN,x.PpL,x.PpH,x.PIDH,x.PIDL,'Lazo abierto')
```



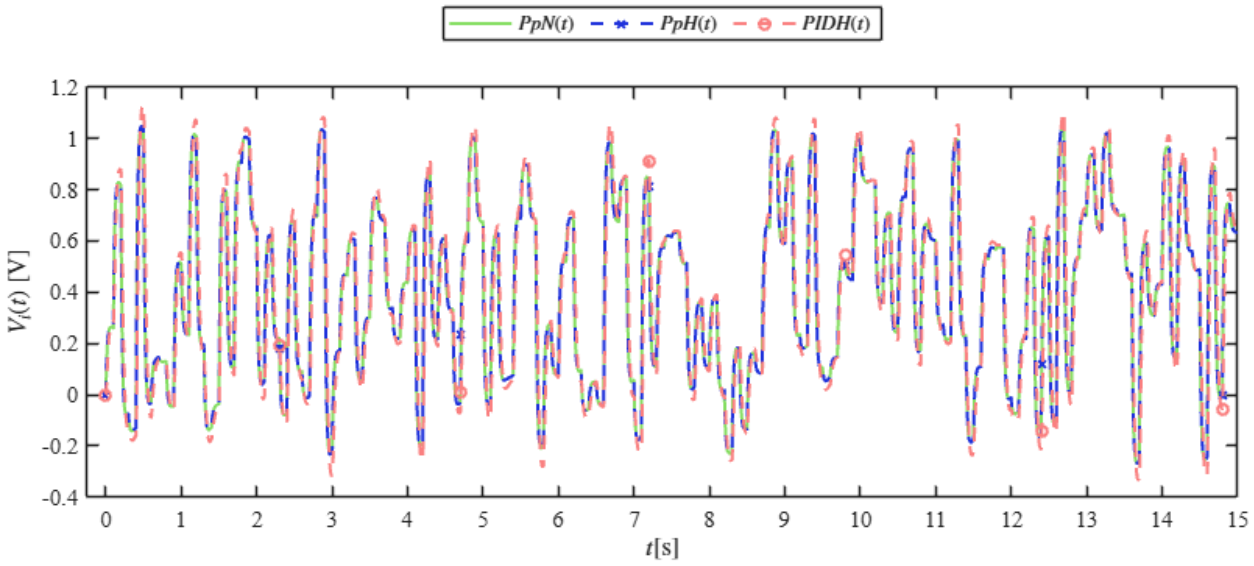
Respuesta al hipotenso

```
plotsignals(x.t,x.PpN,x.PpL,x.PpH,x.PIDH,x.PIDL,'Hipotenso')
```



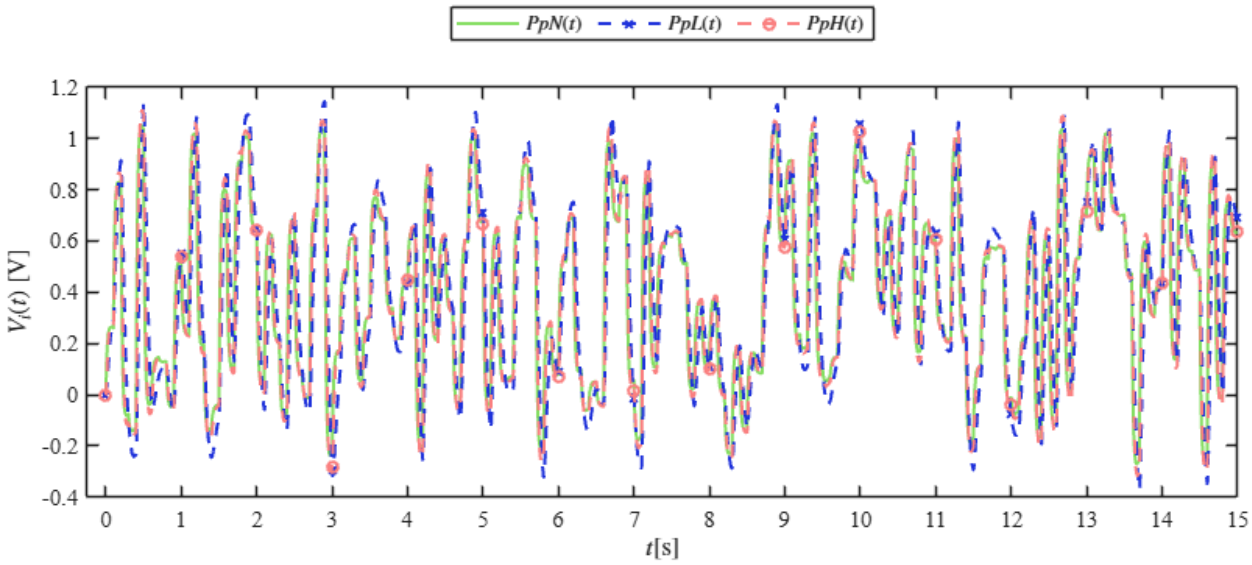
Respuesta del Hipertenso

```
plotsignals(x.t,x.PpN,x.PpL,x.PpH,x.PIDH,x.PIDL,'Hipertenso')
```



Modelo de EDOs

```
par_hipo = [0.010, 0.001, 1.000, 3.500]; %[Z,L,R,C]
par_normo =[0.010, 0.001, 1.000, 1.500];
par_hiper = [0.010, 0.001, 1.000, 1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpN,PpL,PpH,'','','individuos');
```




```
function Pp = cardio(par,Pa,t)
 Z = par(1); L = par(2); R = par(3); C = par(4);
 dt = t(2);
 n = round(t(end)/dt);

 FL = zeros(length(t),1); FL(1) = 0;
 Pp = zeros(length(t),1); Pp(1) = 0;

 for i = 1:n
    FL(i+1) = FL(i) + ((-1/L)*Pp(i)+(1/L)*Pa(i))*dt;
    Pp(i+1) = Pp(i) + ((1/C)*FL(i) - (1/(Z*C)+1/(R*C))*Pp(i)+1/
(Z*C)*Pa(i))*dt;

 end


end
```

Funcion: Respuestas de la simulacion

```
function plotsignals (t,PpN,PpL,PpH,PIDL,PIDH,signal)
set(figure(),'color','w')
set(gcf,'units','inches','position',[1,1,25,10])
set(gca,'fontname','times new roman','fontsize',11)
hold on; grid off; box on;
clr1 = [118, 196, 87]/225;
clr2 = [27, 44, 193]/225;
clr3 = [221, 113, 113]/225;
order = [clr1;clr2;clr3]; colororder(order)
```

```
if signal == "individuos"
    plot(t,PpN,'-',t,PpL,'--x',t,PpH,'--
o','LineWidth',1.5,'MarkerSize',5,'MarkerIndices',...
        1:1e3:length(t));
    L = legend('$PpN(t)$','$PpL(t)$','$PpH(t)$');

elseif signal == "Lazo abierto"
    plot(t,PpN,'-',t,PpL,'--x',t,PpH,'--
o','LineWidth',1.5,'MarkerSize',5,'MarkerIndices',...
        1:1e3:length(t));
    L = legend('$PpN(t)$','$PpL(t)$','$PpH(t)$');

elseif signal == "Hipotenso"
    plot(t,PpN,'-',t,PpL,'--x',t,PIDL,'--
o','LineWidth',1.5,'MarkerSize',5,'MarkerIndices',...
        1:1e3:length(t));
    L = legend('$PpN(t)$','$PpL(t)$','$PIDL(t)$');

elseif signal == "Hipertenso"
    plot(t,PpN,'-',t,PpH,'--x',t,PIDH,'--
o','LineWidth',1.5,'MarkerSize',5,'MarkerIndices',...
        1:1e3:length(t));
    L = legend('$PpN(t)$','$PpH(t)$','$PIDH(t)$');

end
set(L,'interpreter','latex','location','northoutside','orientation','horizont
al')

xlabel('$t$[s]','interpreter','latex')
ylabel('$V_i(t)$ [V]','interpreter','latex')
xlim([-0.25 15]); xticks(0:1:15)
ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
exportgraphics(gcf,[signal,'_matlab1.pdf'],'contenttype','vector')
end
```